

0661

$$11. a. r_s = 1 - \frac{6}{n(n^2-1)} \sum_{i=1}^n d_i^2 = 1 - \frac{6}{12(12^2-1)} \times 454 \approx -0.587$$

$$b. H_0: \rho_s \geq 0, H_a: \rho_s < 0$$

$$n=12, \alpha=0.05, CV = 0.497$$

$-0.497 > -0.587 \Rightarrow \text{Reject } H_0, \text{ 負相關}$

$$12. a. r_s = 1 - \frac{6}{r(r^2-1)} \sum_{i=1}^r d_i^2 = 1 - \frac{6}{9(80)} \times 208.5 = -0.1375$$

b. H_0 : 兩資料無關聯

H_a : 有.

$$n=9, \alpha=0.05, CV = 0.6$$

$-0.1375 < -0.6 \Rightarrow \text{Reject } H_0, \text{ 負相關}$